PROPERNESS OF A FINITE TYPE MORPHISM TO THE LIMIT IS EVENTUAL

THE STACKS PROJECT AUTHORS

Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Proper limit of proper fp. Limits, Lemma lemma-proper-limit-of-proper-finite-presentation (same chapter)
- Finite type eventually closed. Limits, Lemma lemma-finite-type-eventually-closed (same chapter)

Source reference. Original lemma 09ZR: sources/stacks-chow/tags.txt, source lines 12609–12609. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 081F: sources/stacks-chow/tags.txt, source lines 10215–10215. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 081B: sources/stacks-chow/tags.txt, source lines 10211–10211. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $X = \lim X_i$ be a directed limit of schemes over S with affine transition morphisms. Let $Y \to X$ be a morphism of schemes over S. If $Y \to X$ is proper, X_i quasi-compact and quasi-separated, and Y locally of finite type over S, then $Y \to X_i$ is proper for i large enough.*

Proof. Choose a closed immersion $Y \to Y'$ with Y' proper and of finite presentation over X, see Lemma 09ZR. Then choose an i and a proper morphism $Y'_i \to X_i$ such that $Y' = X \times_{X_i} Y'_i$. This is possible by Lemmas 01ZM and 081F. Then after replacing i by a larger index we have that $Y \to Y'_i$ is a closed immersion, see Lemma 081B. □

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